

# DANIEL NG

604-727-3233 | [daniel.wnh@outlook.com](mailto:daniel.wnh@outlook.com) | [danielng.pages.dev](http://danielng.pages.dev) | [linkedin.com/in/danielngEE](https://linkedin.com/in/danielngEE)

## Education

### University of British Columbia

BASc in Electrical Engineering (Co-op, available May 2026)

Expected Graduation: May 2028

Vancouver, BC

## Experience

### Norsat International Inc.

Hardware Intern

June 2025 – August 2025

Richmond, BC

- Performed gain, power saturation, 1-dB compression point, spurious emissions, third-order intermodulation distortion, and noise figure characterization on 20+ 40W, 50W, and 80W Ku-Band Block-Up Converters using Keysight PNA-X Vector Network Analyzers, Signal Analyzers, RF jigs, and multimeters to verify hardware parameters.
- Performed RF power calibration by sweeping DAC codes controlling voltage-variable attenuation and PA bias while monitoring output power and stability.
- Assembled and troubleshooted BUCs and LNBs at the PCBA level, performing microscopic SMT and THT soldering and microscopic inspection on MMIC chips to meet production quality standards.
- Verified and troubleshooted SSPA boards by conducting continuity tests and cross-referencing schematics to identify and rectify DC bias instability and RF path discontinuities, ensuring 100% yield for high-power MMIC stages.

## Formula UBC Racing (Electrical & Firmware)

### Project Lead: Timing Gates | *Embedded C, Layout and Schematic Design*

September 2025 – Present

- Leading the full design cycle of a 650 nm laser gate receiver in Altium, mentoring two MechE students to develop optomechanical shrouding coupled with a **custom analog front end** (photodiode+transimpedance amplifier (TIA)+Schmitt trigger) to maintain 3.3V CMOS logic and 2.4GHz communication between master-slave ESP32-S3 MCUs.
- Characterizing baseline photodiode thermal dark current and TIA voltage noise to calibrate an adjustable hysteresis threshold via potentiometer, targeting a 20dB Signal-to-Noise Ratio (SNR) and a Bit Error Rate (BER) of  $< 10^{-7}$  to ensure reliable high-speed data integrity between MCUs.

## Projects

### RISC-V Single Cycle CPU | *SystemVerilog, Altera DE10-Lite FPGA, Quartus Prime, Questa*

- Architected and implemented a Single-cycle RISC-V processor in SystemVerilog with memory-mapped I/O onto Altera's DE10-Lite with functioning register file, ALU, and control unit, supporting a 10-instruction subset of the RISC-V ISA.
- Wrote a self-checking non-synthesizable testbench with assertion-based verification to verify instruction-level correctness and used Questa to trace signals through the synchronous timing datapath.

### Tron Legacy VGA Driver | *C, Altera DE10-Lite FPGA, MMIO, VGA*

- Developed a custom VGA graphics driver using Memory-Mapped I/O (MMIO) to manage frame buffers, rendering dynamic light traces and a gameplay perimeter while maintaining high-speed pixel updates.
- Implemented an interrupt-driven control scheme for human input, utilizing an Interrupt Service Routine (ISR) to handle asynchronous pushbutton events, significantly reducing input latency compared to traditional polling.
- Engineered an autonomous robot opponent with predictive collision logic; the algorithm performs real-time memory reads of the VGA buffer to scan for obstacles and dynamically updates movement vectors to avoid crashes.

### Autonomous Retrieval System | *SolidWorks, Arduino, C/C++, Hand tools*

- Developed embedded C firmware for Arduino Uno to interface the HC-SR04 sensor and SG90 servo motor, enabling **successful** fully autonomous target detection and retrieval given a 15V power constraint.

## Technical Skills

**Digital Design/HDL/Low-Level:** SystemVerilog, RISC-V Assembly, FSM Implementation & Design

**Embedded Systems:** ESP32 and 8051-family MCUs, C Embedded Firmware Development, RV32 Bare Metal Programming, MMIO

**Design Tools:** Quartus Prime, Questa/ModelSim, Altium Designer, CPULator

**Hardware & Lab:** VNA, Signal Analyzer, PCB Design and Rework, SMT/THT Soldering, Oscilloscope, Multimeter